

ADVANCED BIOMETRIC SYSTEMS AND SECURITY (AI461)

Face Recognition Using Local Texture Descriptors

A Literature Critique of Local Binary Patterns (LBP) and Local Derivative Patterns (LDP)

Group Project — Phase I: Literature Review & State-of-the-Art

Department of Artificial Intelligence | SVNIT Surat

Group Members: Pranav sai, Harshit shah

Agenda

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Biometric Face Recognition

Why faces, and why it's a hard problem

02

Paper 1 — LBP

Ahonen, Hadid & Pietikäinen, IEEE TPAMI 2006

03

Paper 2 — LDP

Zhang, Gao, Zhao & Liu, IEEE TIP 2010

04

Comparative Critique

How LDP builds on and differs from LBP

05

Conclusion & References

Summary and what we implement in Phase II

Face Recognition as a Biometric

Face recognition identifies or verifies a person from a digital image, using facial appearance rather than a possession (card) or a secret (password). **It is non-intrusive and widely deployable — but appearance changes constantly.**



Illumination

Lighting direction/intensity changes pixel values drastically



Pose

Head rotation alters the visible facial geometry



Expression

Local deformations around eyes/mouth



Aging

Long-term texture and shape changes (Dup I/II sets)

Two Representation Families

Holistic methods

PCA (Eigenfaces), LDA (Fisherfaces), 2D-PCA — treat the whole face as one vector. Sensitive to pose/illumination misalignment.

Local feature-based methods

Local Feature Analysis, EBGM (Gabor jets), and — the focus of this review — texture-based micropattern descriptors (LBP, LDP). More robust to local variation; the component-based approach has been shown to outperform holistic recognition.

Both papers reviewed here belong to the local, micropattern-based family.

01

PAPER 1

Face Description with Local Binary Patterns

The foundational LBP micropattern descriptor for face recognition

T. Ahonen, A. Hadid, M. Pietikäinen — IEEE TPAMI, Vol. 28, No. 12, Dec. 2006

Data Acquisition & the Basic LBP Operator

Rationale

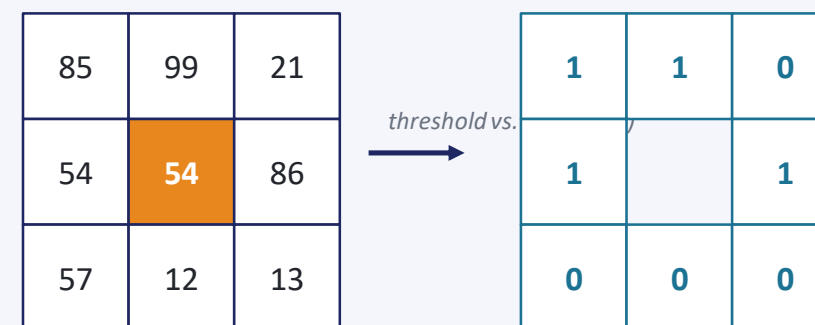
A face is treated as a composition of micropatterns (flat areas, edges, spots) that can be well described by texture operators — no need to build one holistic template.

Dataset — FERET

- 14,051 grayscale images of 1,199 subjects
- fa (1,196) — gallery, frontal, neutral
- fb (1,195) — alternate expression
- fc (194) — different illumination
- Dup I (722) / Dup II (234) — later in time, aging

Only frontal faces are used, normalized/cropped using eye coordinates.

The Basic LBP Operator



Binary (clockwise from top-left): **11001011** → Decimal: **203**

Circular (P,R) neighborhoods generalize this to any radius/point count, with bilinear interpolation. A pattern is **uniform** if it has ≤ 2 bitwise 0/1 transitions — uniform patterns get their own histogram bin (~90% of patterns on FERET faces).

Feature Extraction: The Spatially Enhanced Histogram

Chosen operator: $LBP_u^{2,2}$ computed in 18×21 pixel windows — the best trade-off between recognition performance and feature-vector length (found via extensive parameter search).

1 Divide

The face image is divided into local regions $R_0 \dots R_{\mathbb{N}-1}$ — e.g. 7×7 , 5×5 or 3×3 rectangular windows

2 Describe

An LBP histogram is computed independently within each region

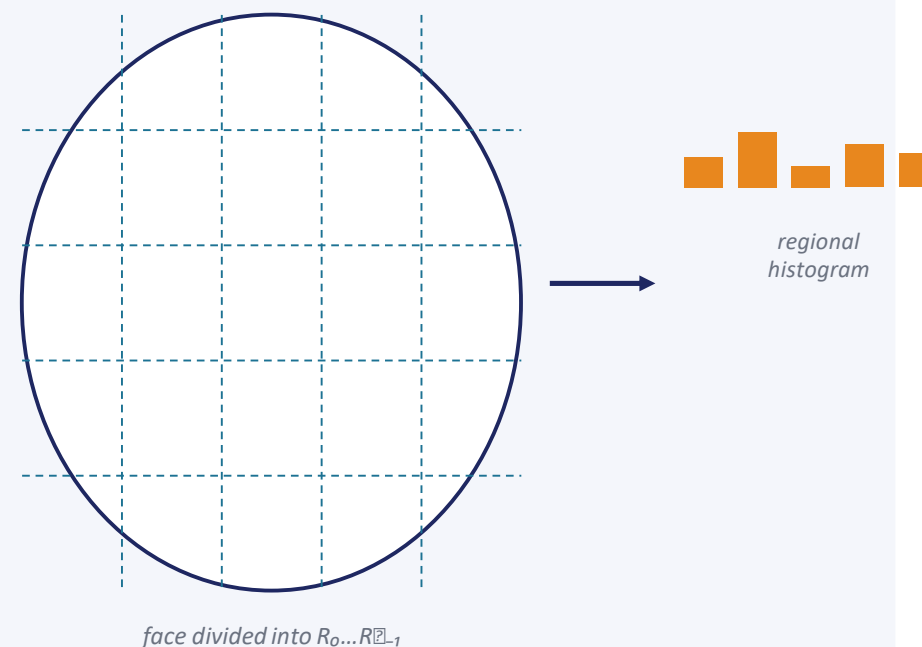
3 Concatenate

The m regional histograms are concatenated into one spatially enhanced histogram of size $m \times n$

4 Weight

Regions are weighted (e.g. eyes weigh more than cheeks), based on the recognition rate each region achieves alone

Regional Division → Global Descriptor



$$H = [H(R_0) \mid H(R_1) \mid \dots \mid H(R_{\mathbb{N}-1})]$$

Three levels of locality in one feature: pixel-level LBP labels → region-level histograms → image-level concatenation.

Classification Scheme

Weighted Chi-Square Distance

$$\chi^2_u(\mathbf{x}, \mathbf{\xi}) = \sum_{ij} w_j \cdot (\mathbf{x}_{ij} - \mathbf{\xi}_{ij})^2 / (\mathbf{x}_{ij} + \mathbf{\xi}_{ij})$$

i, j index the bin and the region; w_j is the learned weight of region j . Nearest-neighbour classification uses this distance between the enhanced histograms of probe and gallery.

Distance measures compared:

- χ^2 statistic outperformed both histogram intersection and log-likelihood distance
- Region weights were assigned from single-region recognition rates — a simple heuristic, not a joint optimisation
- Weighting improved recognition with $P(\text{weighted} > \text{non-weighted}) = 0.976$

Baselines Used for Comparison

PCA

Holistic, Mahalanobis-cosine distance

Bayesian (BIC)

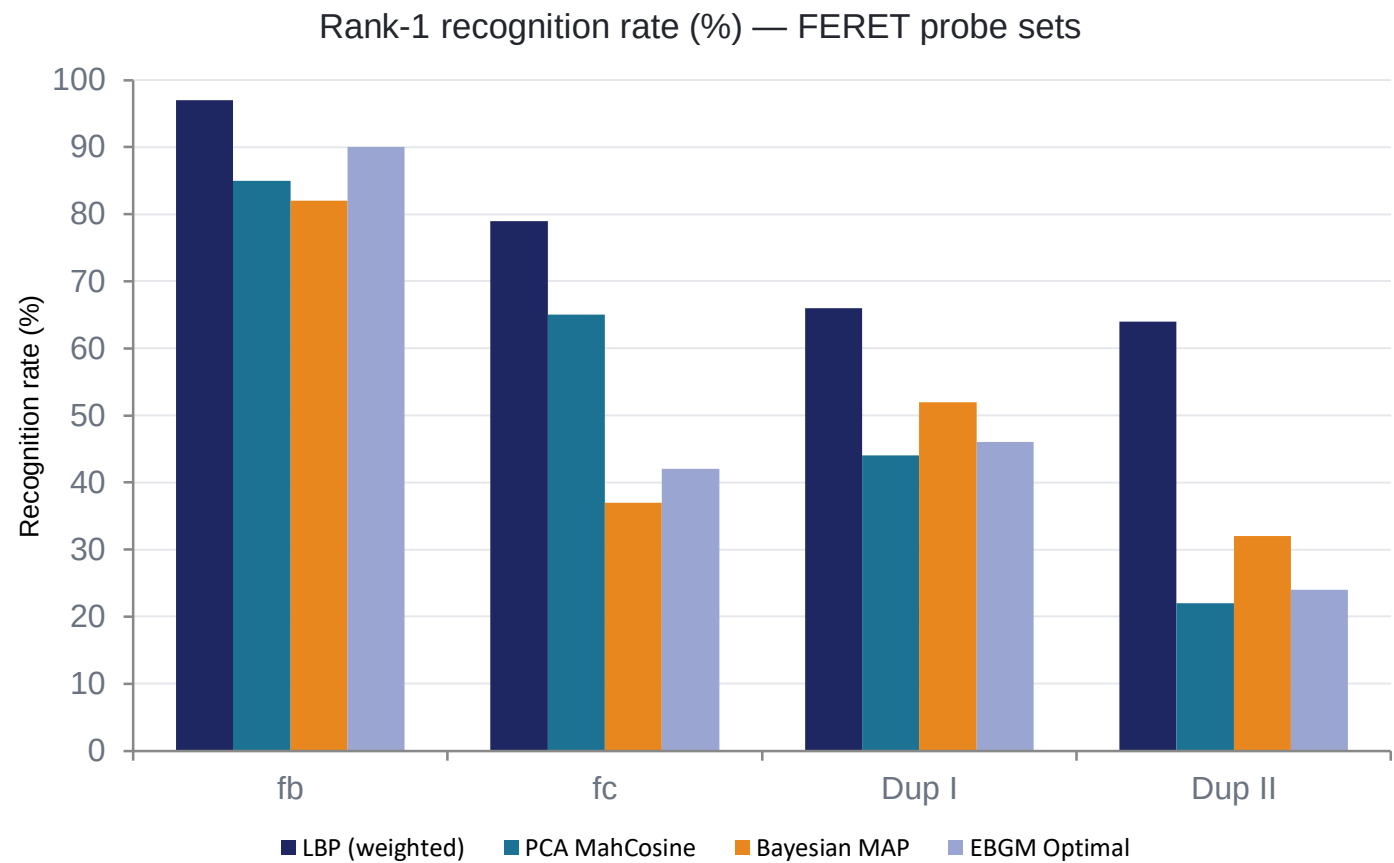
Intra/extrapersonal MAP classifier

EBGM

Gabor jets on a topological face graph

These were run using the CSU Face Identification Evaluation System on the FERET protocol, alongside LBP.

Results: Rank-1 Recognition on FERET



LBP is the only method that survives the fc (lighting-change) set without collapsing.

Weighted LBP clearly beats PCA, Bayesian and EBGM on every probe set, and in the statistical permutation test.

Also from Table 1:
unweighted LBP beat difference-histogram and homogeneous-texture descriptors on every set, and matched/edged out the texton histogram on Dup I/II — confirming the gain is from LBP itself, not just “any texture feature.”

Data reported in Table 2 of Ahonen et al. (2006).

LBP — Strengths & Limitations

Strengths

- ✓ Grayscale-invariant: robust to monotonic illumination changes without any normalisation step
- ✓ Very computationally efficient (simple thresholding + histogramming; ~0.05s per match)
- ✓ No training/learning required for the descriptor itself — directly computed from pixels
- ✓ Spatial histogram preserves coarse geometry that pure texture histograms discard
- ✓ Outperforms PCA, Bayesian and EBGM across all FERET probe sets, including a rigorous statistical test

Limitations

- X Region weights are set by a simple one-region-at-a-time heuristic, not a true joint optimisation — “probably not optimal” in the authors’ own words
- X Encodes only first-order, non-directional gradient information — cannot describe how the gradient direction itself changes across a region
- X Larger local regions gain robustness to face-localisation error, but at the cost of losing fine spatial detail
- X Performance under lighting change (fc) still needs the learned weighting to be competitive

02

PAPER 2

Local Derivative Pattern vs. Local Binary Pattern

A high-order local pattern descriptor that generalises LBP

B. Zhang, Y. Gao, S. Zhao, J. Liu — IEEE Trans. Image Processing, Vol. 19, No. 2, Feb. 2010

Motivation: Beyond First-Order Gradients

The authors reframe LBP as a **nondirectional first-order local pattern**: it thresholds the raw intensity difference between center and neighbour — i.e. the first derivative — and stops there.

LDP instead asks: “does the derivative direction itself change (a turning point), or keep going the same way (monotonic)?” That is a second-order property, and it is captured with a binary co-occurrence code.

Coding function (Eq. 7)

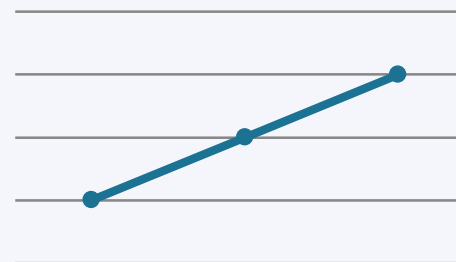
$$f(I'\alpha(Z_0), I'\alpha(Z_i)) = 0 \quad \text{if } I'\alpha(Z_i) \cdot I'\alpha(Z_0) > 0 \quad (\text{same direction})$$

$$f(I'\alpha(Z_0), I'\alpha(Z_i)) = 1 \quad \text{if } I'\alpha(Z_i) \cdot I'\alpha(Z_0) \leq 0 \quad (\text{turning point})$$

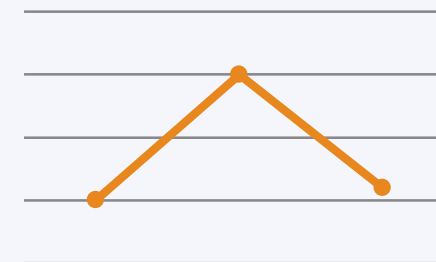
Applied over 16 spatial templates (3- and 4-point) across 4 directions (0°, 45°, 90°, 135°)
→ a 32-bit code per pixel.

Monotonic vs. Turning-Point Patterns

Monotonic → code 0



Turning point → code 1



Second-order LDP

encodes the change of neighbourhood derivative directions — the “turning point” in a given direction — rather than just the raw gradient sign that LBP uses.

n-th order LDP

recursively applies the same coding to the (n-1)-th order derivative image, capturing progressively more detailed directional structure.

General n-th Order LDP & the Gabor Extension



The (n-1)-th order derivative image is recursively fed back in — this is what makes LDP a general, extensible framework rather than a single fixed operator.

General Formulation (Eq. 10–12)

$$\text{LDP}^n(\mathbf{Z}) = \{ \text{LDP}^n \alpha(\mathbf{Z}) \mid \alpha = 0^\circ, 45^\circ, 90^\circ, 135^\circ \}$$

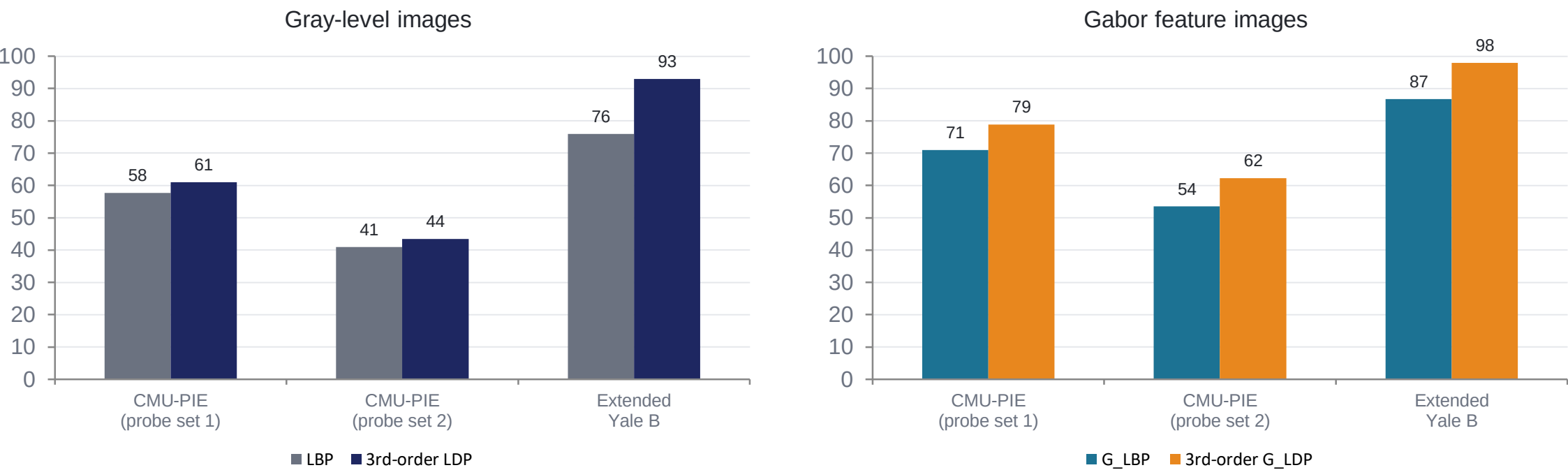
Each direction contributes an 8-bit code from 4 templates $\times$ 2 reference points
→ a 32-bit LDP per pixel, modelled as a spatial histogram exactly like LBP's HLBP.

Extension to the Gabor Feature Domain

The same coding is applied to the real and imaginary parts of Gabor-filtered feature images (G_LDP / G_LBP), alternating real/imaginary across the 4 directions to control code length.

Gabor features add scale/orientation selectivity, which the experiments show consistently boosts both LBP and LDP.

Results: LDP Consistently Outperforms LBP



Rank-1 identification rate (%). Values as reported in the paper's CMU-PIE and Extended Yale B experiments.

3rd order

peaks recognition on FERET, CAS-PEAL, CMU-PIE, Yale B & FRGC
— 4th order overfits to noise and drops back down

+13.7 to 15.0%

higher accuracy than LBP under Gaussian noise σ up to 3, before degrading faster at high noise

0.180s vs 0.054s

per-match cost of 3rd-order LDP vs LBP — roughly 3.3× more computation

LDP — Strengths & Limitations

Strengths

- ✓ Encodes spatial/directional relationships, not just a raw gradient sign — a strictly richer descriptor family than LBP
- ✓ Validated on five databases (FERET, CAS-PEAL, CMU-PIE, Extended Yale B, FRGC) spanning lighting, pose, expression, aging and occlusion
- ✓ The last-order coarse-transition coding curbs noise sensitivity that a naive high-order operator would otherwise suffer
- ✓ Directly compatible with Gabor features (G_LDP), compounding gains from both descriptors
- ✓ No training required, same histogram-intersection matching as LBP — easy to drop in as a replacement

Limitations

- X ~3.3× slower per match than LBP (0.180s vs 0.054s) — a real cost for large-scale or real-time systems
- X Order is a hyperparameter that must be tuned per dataset; performance degrades past 3rd order as noise dominates identity information
- X G_LDP dimensionality is driven mainly by the Gabor filter bank, not LDP itself — needs a reduction step (e.g. LDA) for practical use
- X As a purely hand-crafted, non-learning descriptor, it cannot adapt its templates to a specific dataset the way a learned filter bank could

How LDP Improves on LBP: The Same Idea, Extended

Both papers share the same skeleton — pixel-level binary micropattern → spatial histogram → nearest-neighbour matching. LDP's contribution is entirely in what the binary code encodes.

Aspect	LBP (Ahonen et al., 2006)	LDP (Zhang et al., 2010)
What is encoded	Sign of the raw intensity gradient vs. the center pixel (1st-order, non-directional)	Co-occurrence / turning-point of gradient directions across neighbours (2nd–n-th order, directional)
Code length / pixel	8 bits (P=8 neighbours)	32 bits (4 directions × 8 bits)
Feature domain	Grayscale image only	Grayscale image AND Gabor real/imaginary features (G_LDP)
Datasets evaluated	FERET only	FERET, CAS-PEAL, CMU-PIE, Extended Yale B, FRGC
Best reported result	97% (fb), 79% (fc) on FERET	92.9% (Yale B), 97.9% with Gabor (G_LDP), consistently > LBP
Compute cost	0.054 s / match (fastest)	0.180 s / match (~3.3× LBP)

CONCLUSION

Takeaways for Phase II

- LBP established that micropattern histograms beat holistic PCA/Bayesian/EBGM for face recognition, at very low computational cost.
- LDP is a direct, well-motivated generalisation: encoding directional derivative transitions (not just gradient sign) yields consistent gains across five independent databases.
- The 3rd order is the practical sweet spot for LDP — it captures more detail than LBP without the noise blow-up seen at 4th order.
- For Phase II, we will implement LBP and 2nd/3rd-order LDP on gray-level images, compare rank-1 accuracy and ROC/CMC curves on a shared probe set, and report the FAR/FRR trade-off as required.

Key References

- T. Ahonen, A. Hadid, M. Pietikäinen, “Face Description with Local Binary Patterns: Application to Face Recognition,” IEEE TPAMI, 28(12), 2006.
- B. Zhang, Y. Gao, S. Zhao, J. Liu, “Local Derivative Pattern versus Local Binary Pattern: Face Recognition with High-Order Local Pattern Descriptor,” IEEE TIP, 19(2), 2010.
- P. J. Phillips et al., “The FERET Evaluation Methodology for Face-Recognition Algorithms,” IEEE TPAMI, 22(10), 2000.
- M. Turk, A. Pentland, “Eigenfaces for Recognition,” J. Cognitive Neuroscience, 3(1), 1991.

Thank you — Questions?